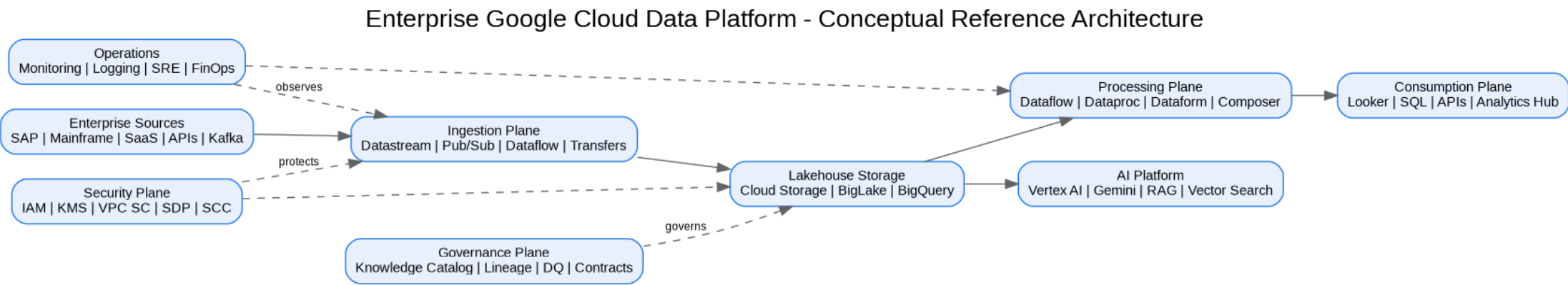


Enterprise Google Cloud Data Platform Blueprint

Executive briefing deck - implementation starter



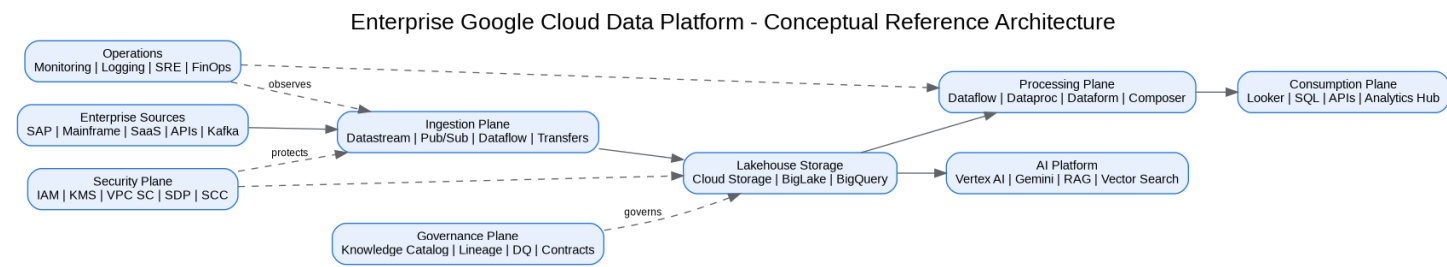
Why this architecture

Modernize fragmented legacy estate

Establish governed data products

Enable analytics, AI and GenAI at scale

Reduce risk through Zero Trust and automation



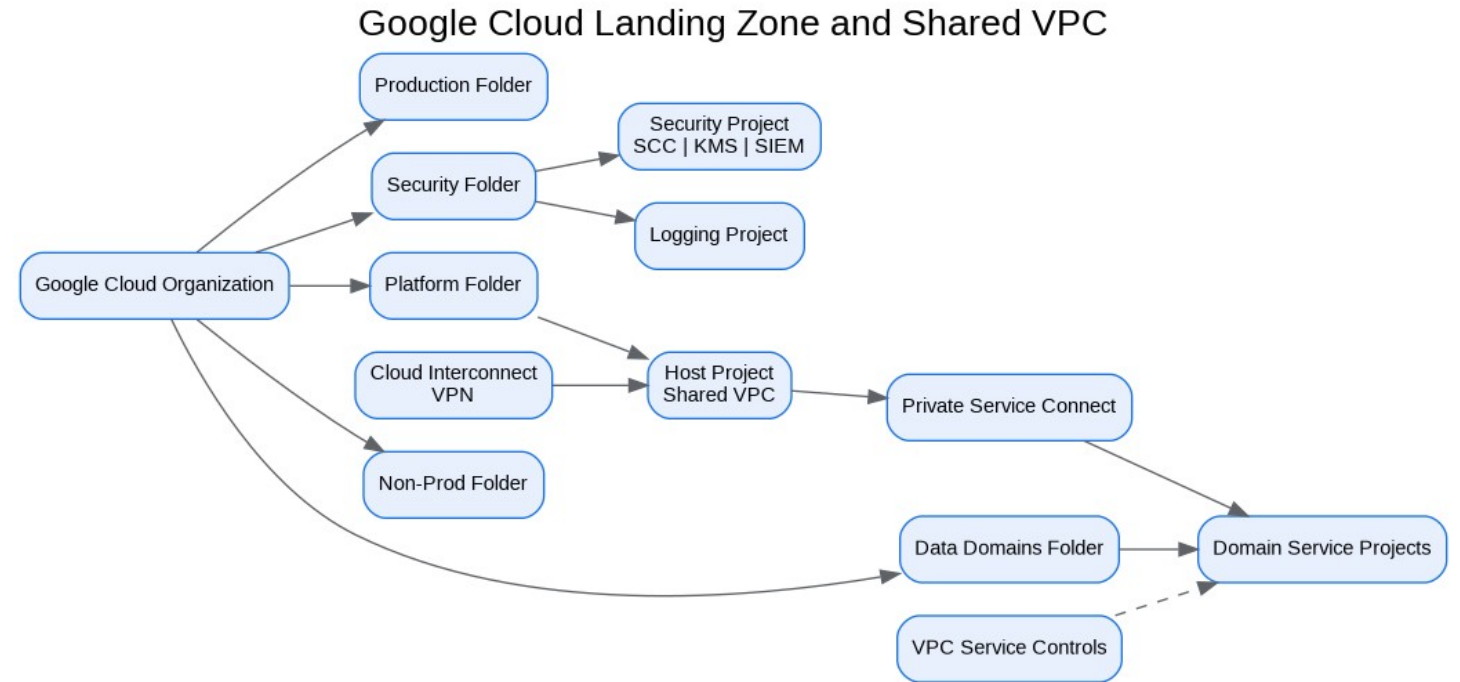
Landing zone foundation

Organization, folders and projects

Shared VPC and private connectivity

IAM, KMS, VPC SC and SCC

Central logging and monitoring



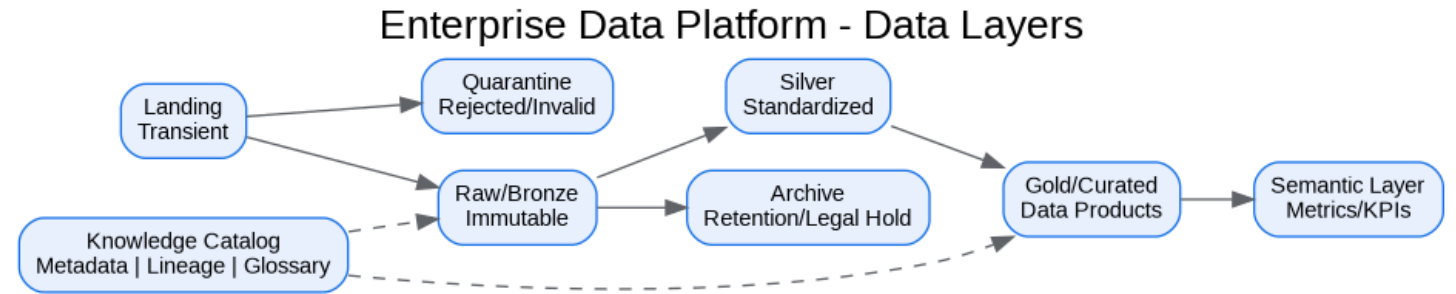
Enterprise data platform

Landing, raw, silver, gold and semantic layers

Batch, CDC, streaming and API ingestion

Knowledge Catalog governance

BigQuery and BigLake serving model



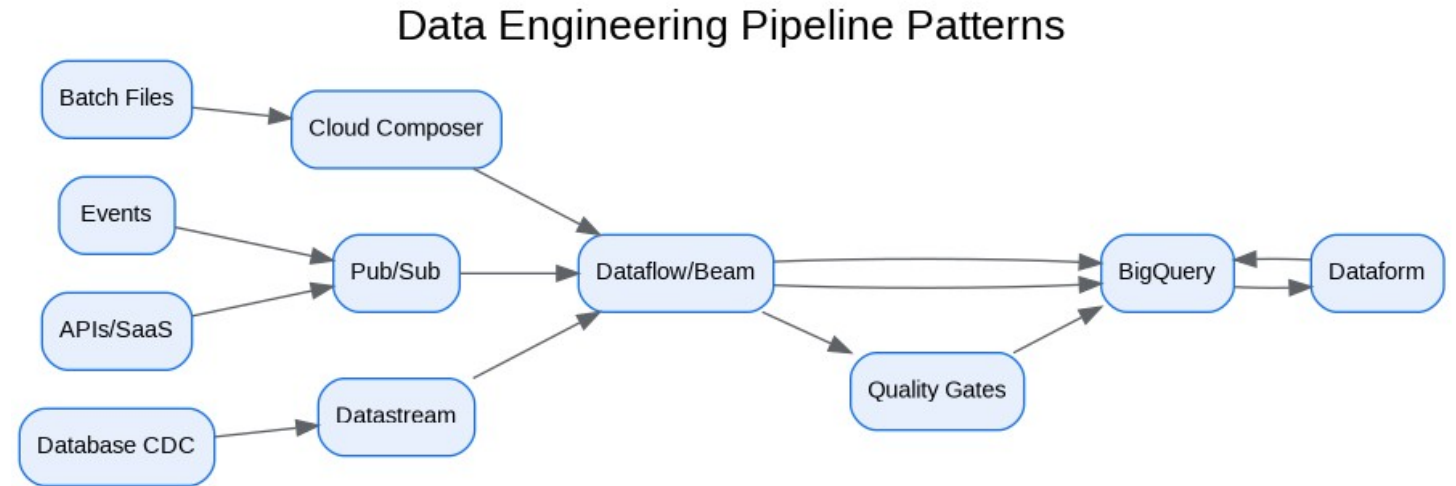
Data engineering standards

Reusable pipeline templates

Dataflow, Beam, Dataproc and Dataform

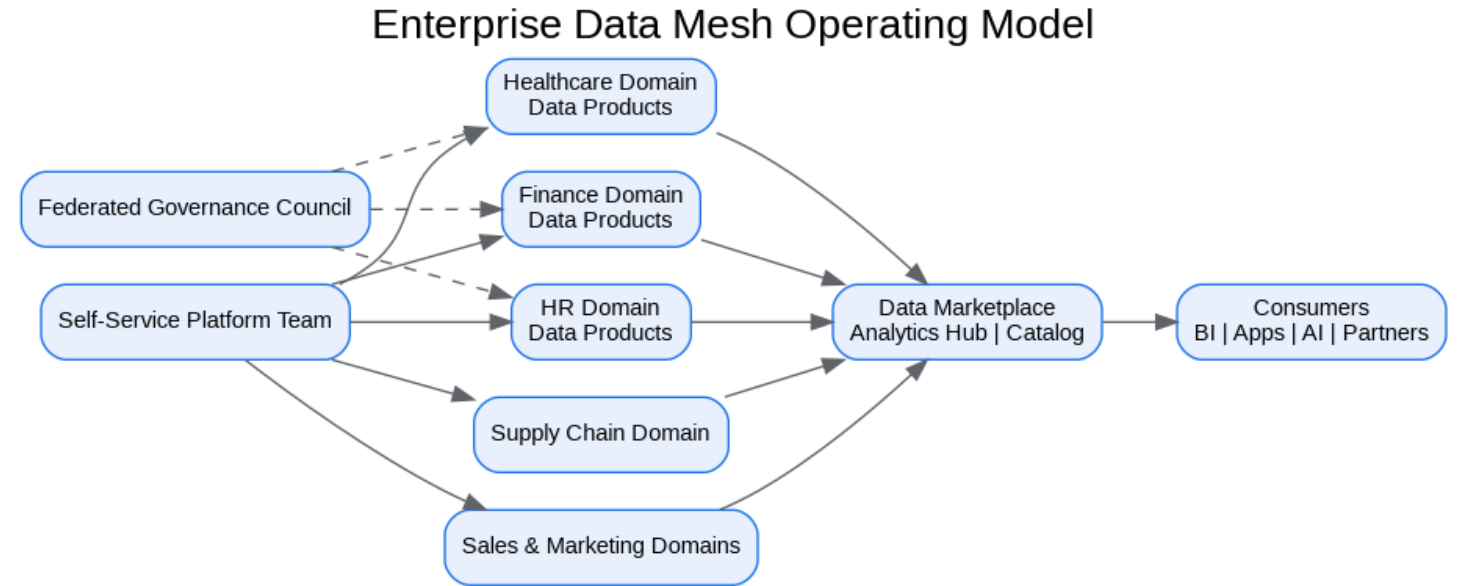
Error handling, DLQ and reconciliation

Observable SLO-driven operations



Data mesh operating model

Central self-service platform
Federated domain ownership
Certified data products
Marketplace and governed sharing



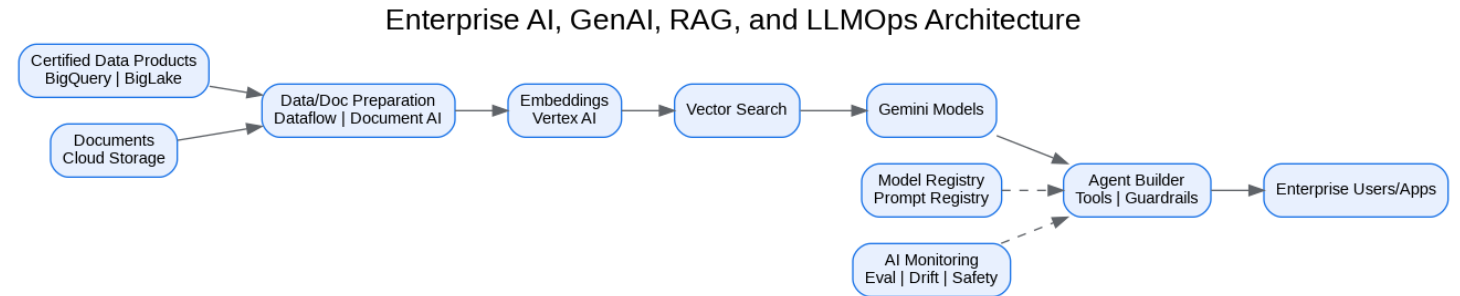
AI-ready architecture

Vertex AI, Gemini and RAG

Vector Search and grounded responses

Prompt/model governance

Sensitive-data and tool-use controls



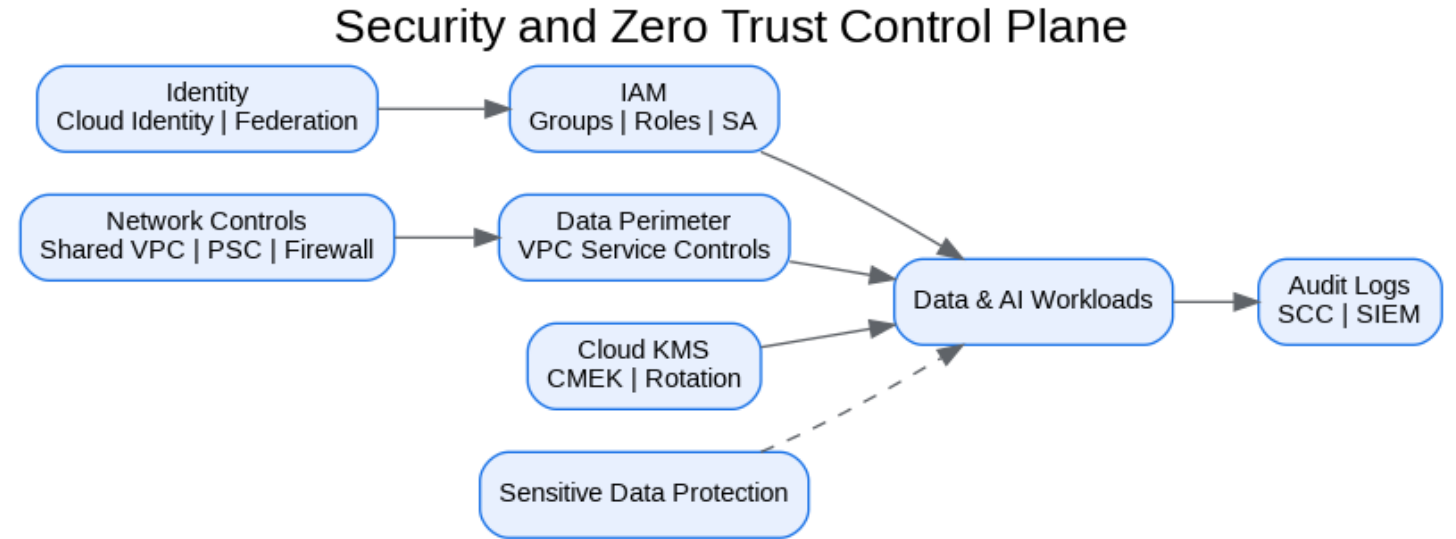
Security and Zero Trust

Identity, network, data and workload perimeters

CMEK for restricted workloads

Sensitive Data Protection and masking

Audit logging and SCC integration



DevSecOps and automation

Terraform golden modules

GitOps and policy-as-code

Promotion gates and approvals

Immutable artifacts and rollback



FinOps and operations

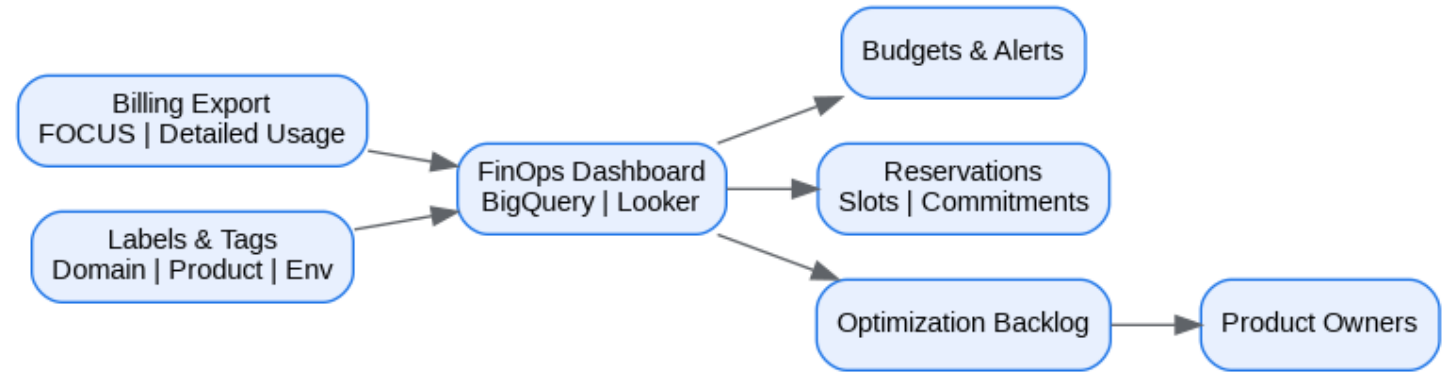
Cost allocation by domain/product

Budgets, reservations and optimization

SLIs, SLOs, runbooks and alerts

DR and production readiness

FinOps Cost Governance Model



Recommended next actions

Confirm executive sponsor and platform product owner

Run architecture foundation workshop

Approve landing-zone MVP scope

Select first production domain

Build reference pipelines and Terraform modules

Schedule security, governance and FinOps reviews